

MULTI-PHASE VARIABLE RELUCTANCE ENERGY HARVESTER FOR SMART BEARING HUB UNITS

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ABSTRACT

This paper reports on the design, optimization, and evaluation of a multi-phase variable reluctance energy harvester (MP-VREH) for integration into a smart bearing hub unit of large commercial vehicles. The MP-VREH converts rotational kinetic energy into electrical energy to enable the supply of smart electronic systems to monitor the bearing condition. The proposed design consists of a ferromagnetic toothed wheel and six identical m-shaped pickup units. Adhering to the space limitations of an example bearing hub unit, critical parameters, such as the pickup unit pole-piece structure, the magnet height, and the number of teeth, have been optimized using 3D finite element analysis. The optimized harvester provides output powers ranging from 0.46 W to 4.77 W at relevant rotational speeds of 100 to 400 RPM. This is deemed to provide sufficient energy for typical application scenarios.

KEYWORDS

Rotational Energy Harvesting, Electromagnetic Transducers, Variable Reluctance, Smart Bearings.

INTRODUCTION

With the rapid advancement of wireless communication technology, an increasing number of sensors are deployed in industrial, structural, and transport equipment to monitor the operational states of critical components. The energy supply has become a significant challenge for maintaining the continuous operation of these sensor systems. In comparison to battery-powered or wired solutions, energy harvesting has the advantage of installation flexibility while also minimizing maintenance for long term operation.

Bearings are critical components used in a multitude of applications and are frequently the source of unplanned maintenance. This motivates fault diagnosis and condition monitoring systems that enable predictive and preventive maintenance. Previous work on energy-autonomous smart bearings has focused on integrating the energy harvester into the bearing [1]. In [2], for example, the authors integrated an electromagnetic energy harvester using Halbach arrays into a bearing structure. They were able to generate up to 131 mW at a rotational speed of 1000 RPM. In [3], a piezoelectric transducer was integrated into a bearing. However, the maximum output power was limited to 2.25 mW at high rotational speeds of several thousand RPM. A challenge in these approaches is that the harvester must be integrated into the bearing, which requires a redesign of the bearing itself and prohibits retrofitting.

Instead, in this work, we propose and optimize a variable reluctance energy harvester (VREH) for integration between two bearings in a bearing hub unit. This simplifies installation and allows for the energy harvester to be retrofitted to existing hub units more easily.

Unlike conventional electromagnetic energy

harvesters, variable reluctance energy harvesters (VREH) exploit the change in magnetic flux caused by variations in the magnetic reluctance within the magnetic circuit [4-6]. Through continuous optimization of the magnetic circuit, it is possible to alleviate the phenomenon of magnetic leakage, thereby enhancing power performance. Xu et al. [7,8] compared six different VREH structures systematically and proposed and optimized an innovative m-shaped VREH, which have enabled energy-autonomous speed sensors [9,10]. The proposed harvester was evaluated under low-speed rotational conditions, generating 480.67 μ W at 10 RPM. A similar e-shaped VREH was exploited by Gong et al. [11] in smart railway axle bearings. Zhang et al. [12] proposed an enhanced VREH, integrating magnetic teeth in the stator to improve the energy density of the harvester. Xu et al. [13] also evaluated the usage of a multi-phase VREH to reduce the cogging torque in the system.

The harvester in this work is based on the m-shaped VREH proposed by Xu et al. However, we propose a further optimization of the pole piece structure in the pickup unit to improve its output power. Moreover, we integrate six pickup units into a multi-phase design, utilizing the constraint space between the two bearings in the bearing hub unit. We also optimize key structural parameters, including the number of teeth and the magnet height, leading to an output power of 0.46 W to 4.77 W at rotational speeds of 100 and 400 RPM, respectively.

These contributions demonstrate a significant improvement in comparison to the state of the art and enable smart bearing hub units powered by the system's continuous rotation.

STRUCTURAL DESIGN AND CONSTRAINTS

A rotational VREH typically consists of a ferromagnetic toothed wheel and a pickup unit containing permanent magnets and a pickup coil [7]. A relative motion between these two components leads to a varying air-gap size in the magnetic circuit and thus a change in the magnetic reluctance. Based on Faraday's law, thus an alternating electrical voltage is induced in the pickup coil, which leads to an electrical power dissipated in the electrical load.

According to [8], the electrical power in an impedance-matched load can be described as

$$P_L = \frac{N_{Coil}^2}{8R_{Coil}} \Delta\phi^2 \left[\frac{\omega N_{Tooth}}{2\pi} \right]^2 \quad (1)$$

Equation (1) shows that the output power depends on several factors, including coil parameters such as the number of turns N_{Coil} and resistance R_{Coil} , the rotational speed ω , the number of teeth in the toothed wheel N_{Tooth} , and the change in magnetic flux $\Delta\phi$. Of these, $\Delta\phi$ is the most complex parameter as it depends on the VREH's physical dimensions and material properties.

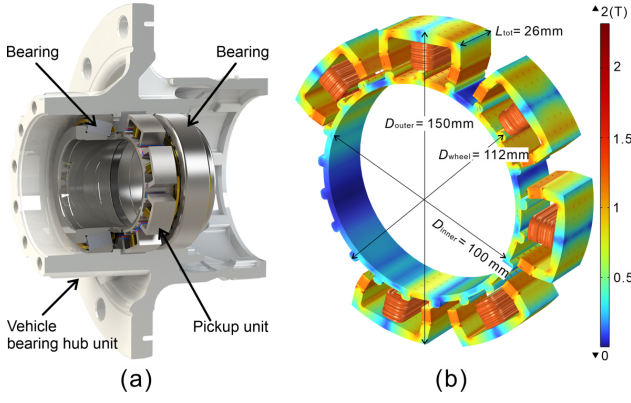


Figure 1: Overview of the proposed MP-VREH; (a) rendering of the harvester integrated into the bearing hub unit; (b) 3D simulation model of the MP-VREH design.

Application use case

For this study, the use case of a vehicular bearing hub unit is employed. This application use case determines the operating conditions and size limitations of the energy harvester. The targeted hub bearing unit is utilized in large commercial vehicles, such as trucks and busses, which limits the typical rotational speeds to a maximum of approximately 400-450 RPM, which is equivalent to 80-90 km/h.

The placement of the VREH in between two cylindrical roller bearings within the hub, limits the available physical space. Consequently, the developed VREH is limited to an outer diameter no greater than 150 mm, an inner diameter of at least 100 mm, and a thickness not exceeding 26 mm. A rendering of the targeted bearing hub unit, including bearings and VREH, is depicted in Fig. 1a.

Multi-phase Variable Reluctance Energy Harvester

The proposed multi-phase variable reluctance energy harvester (MP-VREH) contains a ferromagnetic toothed wheel constructed of electrical steel (NSC-35H230) to be mounted on the inner race of the bearings, as well as six identical m-shaped pickup units mounted on the cast-iron hub. The pickup units are an enhanced version of the pickup structure proposed in [8]. Each pickup unit contains two permanent magnets of NdFeB-N50, a hollow rectangular copper coil, and an m-shaped ferromagnetic pole-piece.

The structure of the MP-VREH with its overall dimensions can be seen in Fig. 1b. The dimensions of each component have been selected to fit the size limitations prescribed by the application use case. Moreover, the six pickup units are arranged in an optimal angular distance to minimize the cogging torque generated by the magnetic interaction of the permanent magnets and the toothed wheel, as introduced in [13].

OPTIMIZATION GOALS AND METHOD

To maximize the output power of the MP-VREH while staying within the physical limitation of the use case, several structural dimensions require optimization. These parameters include the structure of the pickup unit, its magnet height, as well as the number of teeth on the toothed wheel. For all optimizations, a constant pole width and

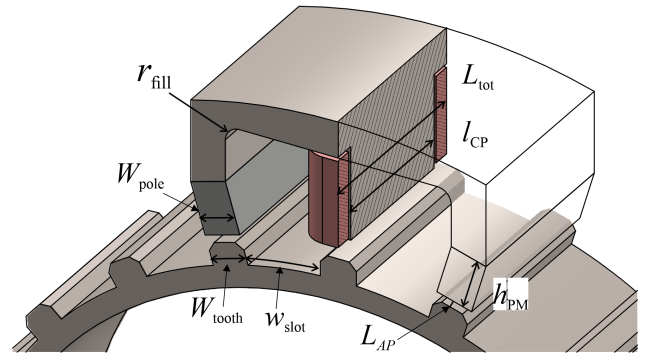


Figure 2: Structure and key dimensional parameters of the pickup unit in the MP-VREH. Capital letters indicate constant values, whereas small letters represent variables.

tooth width is employed.

To study the impact of structural dimensions, a 3D model of the MP-VREH has been implemented in COMSOL Multiphysics to conduct a finite element time-dependent analysis. An example of this model is depicted in Fig. 1b. The simulations utilize the Rotating Machinery Magnetic (rmm) interface to evaluate the model under constant angular velocities ranging from 10.472 rad/s to 41.888 rad/s (approx. 100 to 400 RPM). The Electrical Circuit interface (cir) is used to obtain the electrical output power for a matched resistive load.

Enhanced pickup unit structure

The original m-shaped VREH introduced in [8] utilized a constant pole-piece depth. As a result, the copper coil extends beyond the pole-piece and defines the total depth of the harvester. However, this is not optimal, as it limits the area of the coil that is exposed to flux change and increases the reluctance of the pole-piece. The pickup unit in the MP-VREH therefore contains different pole-piece depths for the center pole, which hosts the coil, and the rest of the pole-piece. As such, the pole-piece covers the coil in the proposed design entirely and has in total a larger depth (i.e., 26 mm in the targeted use case). A close-up view of the proposed pickup unit structure in the MP-VREH and its key parameters are shown in Fig. 2. The center pole l_{CP} needs to be smaller than the total depth L_{tot} to leave space for the coil. A smaller l_{CP} leads to a larger coil area, which increases the number of coil windings. However, it limits the ferromagnetic material in the core, and thus reduces the inductance. A larger l_{CP} has the opposite effects. To identify the optimal value for l_{CP} , a parametric sweep was conducted from 4 mm to 24 mm.

In addition, the pickup unit contains two permanent magnets with height h_{PM} at the end of the outer poles. The permanent magnets are the source of the magnetic field in the VREH. Intuitively, an increased height of the magnet h_{PM} may increase the field strength, and thus provide larger power. However, in a VREH, the output power is determined by the magnetic flux path through the magnets, air gap, and ferromagnetic materials. Since the permeability of permanent magnets is close to that of air, a magnet that is too thin produces insufficient flux, while one that is too thick provides a too large air gap reluctance. This creates a trade-off situation in the circuit, and thus necessitates an optimal h_{PM} to be found. To identify the

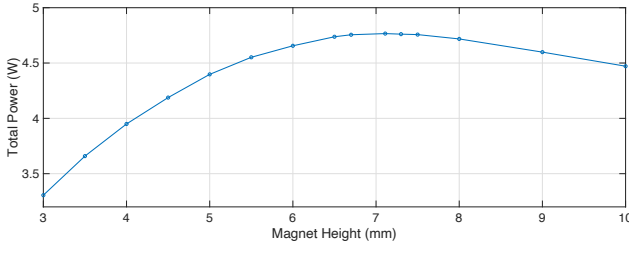


Figure 3: Parametric sweep results (at 400 RPM) of the magnet height in the MP-VREH, indicating an optimum at 7.1 mm.

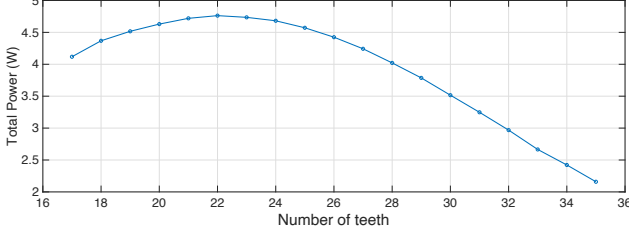


Figure 4: Parametric sweep results (at 400 RPM) of the number of teeth in the MP-VREH, demonstrating highest output power with 22 teeth.

optimal value for h_{PM} in the MP-VREH, a parametric sweep was conducted in COMSOL.

Furthermore, additional minor modifications have been conducted to the original m-shaped pickup, including the optimization of the fillet size r_{fill} .

Optimized toothed wheel

In the original m-shaped VREH, the toothed wheel had the same tooth width (W_{tooth}) and slot width (w_{slot}), and it has not been analyzed what effect an increased slot width has on the flux variation in the VREH. An increased w_{slot} reduces the magnetic flux as the air gap increases, and thus contributes to a larger flux difference. However, the effect of this saturates, while an increasing w_{slot} also reduces the frequency of the generated output voltage. It is thus expected that w_{slot} also has an optimum value.

To identify the optimum slot width, a parametric sweep of the number of teeth in the toothed wheel was conducted. This sweep assumes a fixed value for W_{tooth} , which is equal to the width of the outer poles of the pickup unit W_{pole} . Maintaining six pickup units with an optimal angular distance of 46.36 degrees, the MP-VREH design is limited to a minimum of 17 teeth. The parametric sweep was thus conducted from 17 to 35 teeth.

RESULTS

In this section, the results obtained through COMSOL simulations are presented and discussed. Parametric sweep examples are shown for a rotational speed of 400 RPM.

Pickup unit optimization

The enhanced pickup unit structure of the MP-VREH demonstrates significant output power improvements in relation to the original m-shaped VREH. By widening most of the pole-piece, a significant increase in output power was achieved as compared to the original structure, while maintaining the same total depth. The best result was obtained when l_{CP} was designed to be 20 mm, restricting

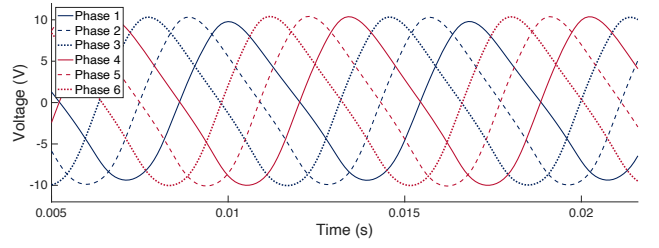


Figure 5: Output voltage of the six pickup units in the MP-VREH at a rotational speed of 400 RPM.

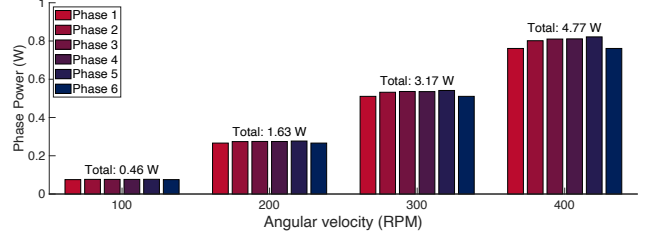


Figure 6: Output power for each phase and total output power of the MP-VREH as a function of rotational speed.

the copper coil to 979 windings with a total coil resistance of 59.84 ohms and an inductance of 10.44 mH. Optimizing the fillet size r_{fill} to a value of 4 mm further increased the output power by 20%. In total, a 3.25-fold improvement in output power was achieved.

Fig. 3 depicts the result of a parametric sweep of the magnet height h_{PM} in the MP-VREH. Based on this parameter sweep, it was determined that the optimal value for h_{PM} is 7.1 mm. Although smaller magnets can be utilized to reduce cost, the output power reduces significantly at the lower range of magnet heights.

Optimized toothed wheel and total output power

Fig. 4 shows the effect of sweeping the number of teeth in the toothed wheel. The result verifies that an optimal number of teeth exists. For the proposed use case and its physical space limitations, the highest output power was achieved with $N_{Tooth} = 22$.

Combining the enhanced pickup structure (incl. an optimal l_{CP}), the optimized magnet height h_{PM} , and the optimized number of teeth, the output voltage and power of the MP-VREH was estimated at different rotational speeds. Average peak output voltages of 3.02 V to 9.75 V have been observed over the speed range of 100-400 RPM. Fig. 5 exemplifies the multi-phase output voltages of the MP-VREH at a rotational speed of 400 RPM. The output power results are depicted in Fig. 6. The figure shows the output power generated for each pickup unit, as well as the total output power for each speed scenario. The generated output power increases with rotational speed from 0.46 W at 100 RPM to 4.77 W at 400 RPM. The increase is approximately linear rather than quadratic, which is due to the resistive matching of the load (i.e., $R_{Coil} = R_L$). This means that the output power could further be increased by a complex impedance matching. However, this is difficult to be attained dynamically in a practical deployment.

It is obvious from the results that Phase 1 and Phase 6 generate less power than the other pickup units. The reason for this is that the six pickup units are not spread evenly

over the toothed wheel. Instead, their arrangement is based on the optimal angular distance to minimize cogging torque. Consequently, there is a larger space between the last and first pickup unit. Neighbouring pickup units interact by linking their magnetic fields, contributing to power generation. Due to the larger angular distance to one of their neighbours, this effect is significantly reduced for the first and last units.

CONCLUSIONS

This paper proposed MP-VREH, a six-phase variable reluctance energy harvester designed to enable smart bearing hub units. Based on an m-shaped VREH, we demonstrated that the output power could be increased significantly by enhancing the pickup unit structure. Moreover, we showed that several critical parameters can be optimized with respect to the output power in the targeted application use case.

The optimized MP-VREH has 22 teeth in the toothed wheel, and a magnet height of 7.1 mm. The center pole of the pole-piece was determined to have an optimal depth of 20 mm. With these parameters, the MP-VREH is capable to generate between 0.46 W and 4.77 W at rotational speeds of 100 RPM to 400 RPM, respectively. These conditions represent typical operating conditions for large commercial vehicles, such as trucks and buses, with velocities equivalent to 20-80 km/h.

With the aim to supply power to a condition monitoring system for the integrated bearings, the obtained power levels are deemed sufficient to provide enough energy for sensing, data processing, and wireless communication in most practical scenarios. Indeed, with modern low-power electronics, it is assumed that these power levels exceed typical application requirements. Consequently, future research activities can focus on reduced cost and environmental impact rather than further increasing output power and may address even more space constraint scenarios.

In conclusion, the proposed MP-VREH demonstrates to be a suitable energy harvesting technology for the targeted use case, enabling the implementation of energy-autonomous smart bearing hub units.

ACKNOWLEDGEMENTS

The research in this work was financially supported by the Chips JU project LoLiPoP-IoT (Long Life Power Platforms for Internet of Things), <https://www.lolipop-iot.eu>, grant no. 101112286 and by the Knowledge Foundation through the NIIT project under grant no. 20180170.

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